

Application of Machine Learning to Understand the Energy use of Collaborative Robots

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Abstract—The use of collaborative robots is ever-growing, which increases the issue of energy use during prolonged operation. The goal of the presented research is the evaluation of the importance of various factors that may have an influence—such as different loads, movement types, and temperatures of the robot system. The information is contained within a publicly available dataset, "Dataset for Energy Assessment of Collaborative Robots" which contains data on the UR3e cobot from various experiments. In the presented research, the authors combine data from multiple experiments to obtain a single dataset containing 226,472 observations, allowing for cross-comparison of different experimental setups. Statistical methods including three-way factorial ANOVA with interaction effects, effect size analysis (eta-squared), post-hoc comparisons (Tukey's HSD), and Random Forest feature importance analysis are applied to determine the influence of selected factors on energy consumption. The main goal of this research is to determine which factors have the largest influence on the energy use of the cobot, which may serve as a starting point for a knowledge base of guidelines and best practices for the general setup, as well as the programming and control of collaborative robots in terms of lowering their energy use. Results revealed that temperature is the dominant predictive factor (80.84% feature importance), followed by movement type (18.91%), while load showed negligible influence (0.26%).

Keywords—collaborative robots, energy use, machine learning

I. INTRODUCTION

II. INTRODUCTION

Collaborative robots (cobots) have become an integral component of modern industrial environments due to their flexibility, ease of deployment, and ability to safely operate alongside human workers. As their adoption continues to grow across manufacturing, logistics, and service-oriented applications, the cumulative energy demand associated with prolonged and repetitive operation has emerged as an increasingly important consideration [1]. Reducing energy consumption is not only relevant from a sustainability and cost perspective, but also directly impacts system efficiency, thermal stability, and long-term operational reliability [2].

While significant research has focused on improving the mechanical design, control accuracy, and safety of collaborative robots, comparatively less attention has been paid to systematically quantifying how operational and environmental factors influence their energy use [3]. Factors such as movement type, payload conditions, and thermal

state are often adjusted empirically during deployment, without clear, data-driven guidelines regarding their relative impact on energy consumption [4]. This lack of quantitative understanding limits the ability to optimize robot operation in energy-sensitive contexts.

In recent years, the availability of high-resolution operational datasets and advances in statistical analysis and machine learning techniques have enabled more detailed investigations into the drivers of energy consumption in robotic systems [5]. By combining classical statistical methods with data-driven models, it becomes possible not only to assess the statistical significance of individual factors, but also to estimate their relative importance and practical relevance.

In this work, we analyze a publicly available dataset containing energy-related measurements of a collaborative robot operating under different movement types, load conditions, and thermal states. By aggregating data across multiple experimental setups and applying factorial analysis of variance alongside machine learning-based feature importance analysis, the study aims to identify which factors most strongly influence energy consumption and whether interaction effects are present. The results are intended to provide a quantitative basis for energy-aware configuration and programming of collaborative robots, serving as an initial step toward the development of practical guidelines for reducing their energy use in real-world applications.

The following research questions are the main focus of the research:

- RQ1 — Do the categorical factors (type of movement, temperature, load) have statistically significant effects on energy consumption?
- RQ2 — What is the magnitude of these effects?
- RQ3 — Do these factors interact with each other to influence energy consumption?
- RQ4 — Which specific factor levels differ significantly from each other?

III. DATASETS

The used dataset is the "Dataset for Energy Assessment of Collaborative Robots" [5], which provides a number of subsets comparing the performance of an UR3e collaborative robot in various environments. Namely, the datasets measure different currents at nominal voltage during linear and joint-driven movement [], movement with load and

without load and cold (movement executed when the robot was started from a cold/room temperature state) and warm movement (movement executed when the robot was already warmed up from previous operations).

The data provided consists of over 20,000 measurement points. The first step is the calculation of energy usage of the robot, as a product of the nominal voltage and current used between two measurements (all of which are provided as parts of the dataset):

$$E_i = I_i \cdot U_i \cdot \Delta t_i \quad (1)$$

The dataset is then filtered to remove any points where energy is zero. Then, three new columns are added, where a binary indicator ("1" for positive, "0" for negative) is used to note if the movement was linear, if the movement was performed under a load, and if the movement was warm. The other columns, which provide further detail on currents and other parameters during movement are removed and not used in further analysis, as the main research goal is to determine the influence of individual factors on the output. The problem arises that now, there is a one-to-many relationship between the energy and the binary input data. To address this, the authors group the data by the input combination, averaging the used energy across all uses of particular combination of factors. This dataset is final, and used in the further analysis. The filtered dataset consists of 6,957 rows. The categorical nature of the inputs result in eight factor combinations.

IV. METHODS OF ANALYSIS

The following methods were used for the evaluation of the dataset:

A. Factorial ANOVA (Analysis of Variance)

A three-way factorial ANOVA (Type III) was conducted to test the main effects of each factor and all possible interaction effects. The null hypothesis states that factor levels have equal mean energy consumption [6]:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k \quad (2)$$

The ANOVA partitions total variance into [7]:

- Main effects: Move Type, Temperature, Load
- Two-way interactions: Move Type $\times$ Temperature, Move Type $\times$ Load, Temperature $\times$ Load
- Three-way interaction: Move Type $\times$ Temperature $\times$ Load
- Residual (error) variance

Statistical significance was assessed using $\alpha = 0.05$.

B. Effect Size Analysis (Eta-squared)

Eta-squared (η^2) quantifies the proportion of total variance explained by each factor [8]:

$$\eta^2 = \frac{SS_{effect}}{SS_{total}} \quad (3)$$

where SS_{effect} is the sum of squares for the effect and SS_{total} is the total sum of squares. Effect sizes are interpreted as:

- Small: $\eta^2 < 0.06$
- Medium: $0.06 \leq \eta^2 < 0.14$
- Large: $\eta^2 \geq 0.14$

C. Interaction Plots

Interaction plots were used to visualize how the effect of one factor depends on the levels of another factor. In these plots, the response mean is shown as a function of one factor, with separate lines corresponding to different levels of a second factor, providing an intuitive graphical assessment of potential interaction effects.

Formally, an interaction is present when the relationship between a factor A and the response variable differs across levels of a factor B . This is reflected in non-parallel lines in the interaction plot, indicating that the marginal effect of A is not constant with respect to B . Conversely, approximately parallel lines suggest additive effects and the absence of a meaningful interaction [9].

D. Post-hoc Tests (Tukey's HSD)

Tukey's Honestly Significant Difference (HSD) test was employed to conduct exhaustive pairwise comparisons among all levels of the considered factors while rigorously controlling the family-wise error rate. Following a statistically significant omnibus test (typically an analysis of variance), Tukey's HSD evaluates the differences between all possible pairs of group means using a single critical threshold, thereby avoiding the inflation of Type I error that would arise from multiple uncorrected comparisons. This makes the procedure particularly suitable for balanced or approximately balanced designs in which the objective is to identify specific group-level differences rather than merely establishing the presence of an overall effect [10].

Formally, let $\bar{y}_i$ and $\bar{y}_j$ denote the sample means of groups i and j , each estimated from n observations, and let MSE denote the mean squared error obtained from the ANOVA model with ν residual degrees of freedom. The Tukey HSD criterion declares the difference between groups i and j to be statistically significant if

$$|\bar{y}_i - \bar{y}_j| > q_\alpha(k, \nu) \sqrt{\frac{\text{MSE}}{n}},$$

where k is the number of groups being compared and $q_\alpha(k, \nu)$ is the critical value of the Studentized range distribution at significance level α . This formulation ensures that the probability of making at least one false rejection across all pairwise comparisons is bounded by α [11].

By relying on the Studentized range distribution, Tukey's HSD provides a coherent framework for interpreting multiple comparisons within a single inferential procedure. As a result, the test not only identifies which specific pairs of factor levels differ significantly, but also yields confidence intervals for the mean differences that are

adjusted for multiple testing. Consequently, Tukey's HSD offers a robust and interpretable post-hoc analysis, enabling precise attribution of observed effects to particular group contrasts while maintaining strict control over the overall error rate.

E. Random Forest Feature Importance Analysis

To complement the traditional ANOVA approach with a machine learning perspective, a Random Forest Regressor was trained to predict energy consumption based on the three categorical factors. Random Forests are ensemble learning methods that automatically capture non-linear relationships and complex interactions [12].

Model was configured using the following hyperparameters [13]:

- Algorithm: Random Forest Regressor with 200 trees
- Maximum depth: 10 levels
- Minimum samples per split: 5
- Cross-validation: 5-fold
- Performance metric: R^2 (coefficient of determination)

V. RESULTS AND DISCUSSION

A. Descriptive Statistics

Table I presents summary statistics [14] for all eight factor combinations.

The table highlights several key observations. First, the experimental design is well balanced, with sample sizes ranging from 806 to 914 per group. Second, the variability is substantially reduced, as standard deviations fall between 4 and 13, in stark contrast to the full dataset where some groups exhibited standard deviations exceeding 1600. Third, a clear effect of Move Type is evident, with Move Type = 0 consistently associated with higher energy consumption (approximately 8–10) than Move Type = 1 (approximately 5–7). Finally, the effect of Load is consistent across all combinations, with Load = 1 systematically resulting in higher energy consumption than Load = 0.

B. Factorial ANOVA Results

Table II presents the ANOVA results.

The results indicate that both Move Type and Load have a highly significant effect on the outcome, with F values of 48.08 and 16.01, respectively ($p < 0.001$ in both cases). In contrast, Temperature does not show a statistically significant effect ($F = 0.69$, $p = 0.405$). Moreover, none of the interaction terms reached statistical significance, suggesting that the effects of the factors act largely independently.

C. Effect Size Analysis

Table III shows eta-squared values for all factors.

TABLE III
EFFECT SIZES (ETA-SQUARED) FOR ALL FACTORS (FILTERED DATA)

Factor	η^2	Percentage	Interpretation
Move Type	0.00622	0.622%	Small
Load	0.00207	0.207%	Small
Temperature $\times$ Load	0.00011	0.011%	Small
Temperature	0.00009	0.009%	Small
3-way interaction	0.00004	0.004%	Small
Move Type $\times$ Temperature	0.00003	0.003%	Small
Move Type $\times$ Load	0.000003	0.0003%	Small

Figure 1 visualizes these effect sizes.

The effect size analysis shows that Move Type has the largest influence, explaining 0.62% of the variance, followed by Load with 0.21%. In contrast, Temperature contributes a negligible amount of explained variance (0.009%). Taken together, the main effects account for approximately 0.83% of the total variance. Although some factors are statistically significant, the corresponding effect sizes are small by conventional standards, remaining well below the 6% threshold typically associated with even small effects.

D. Interaction Plots

Figure 2 displays interaction plots for all two-way factor combinations.

The interaction plots show that the lines are nearly parallel in all cases, indicating the absence of meaningful interaction effects. For the Move Type $\times$ Temperature comparison, the lines remain almost parallel, confirming the non-significant interaction, with Move Type = 0 consistently exhibiting higher values. In the Move Type $\times$ Load plot, the lines are strongly parallel, indicating that Load increases energy in a similar manner for both move types. Likewise, in the Temperature $\times$ Load plot, the lines are highly parallel, showing that the effect of Load is consistent across temperature levels. Overall, the consistent parallel patterns across all plots corroborate the ANOVA result that no interaction effects are statistically significant.

E. Distribution Analysis

Figure 3 shows energy distributions for each factor level.

The distributions reveal a clear separation for Move Type, with Move Type = 0 consistently shifted toward higher values. In contrast, the Temperature distributions show substantial overlap, in line with its non-significant effect. The effect of Load is visible as a systematic shift, with Load = 1 exhibiting higher values than Load = 0. Extreme outliers above 60 are less prevalent compared to the full dataset, indicating reduced influence of extreme values. Despite this reduction, all distributions remain positively skewed, with a longer tail toward higher values.

F. Post-hoc Tests (Tukey's HSD)

Table IV presents pairwise comparisons.

The results show that Move Type has a significant effect, with Move Type = 1 consuming 3.04 units less energy

TABLE I
DESCRIPTIVE STATISTICS OF ENERGY CONSUMPTION (FILTERED DATA)

Move Type	Temperature	Load	N	Mean Energy	Std. Dev.
0	0	0	806	8.43	9.92
0	0	1	914	10.16	13.18
0	1	0	845	8.06	10.31
0	1	1	821	10.36	12.90
1	0	0	887	5.42	4.30
1	0	1	888	7.05	5.70
1	1	0	892	5.36	4.08
1	1	1	904	7.07	5.75

TABLE II
THREE-WAY FACTORIAL ANOVA RESULTS (FILTERED DATA, TYPE III)

Source	Sum of Squares	df	F	p-value	Significance
Move Type	3833.09	1	48.08	<0.001	***
Temperature	55.28	1	0.69	0.405	ns
Load	1276.54	1	16.01	<0.001	***
Move Type × Temperature	20.89	1	0.26	0.609	ns
Move Type × Load	1.90	1	0.02	0.877	ns
Temperature × Load	68.22	1	0.86	0.355	ns
3-way interaction	26.62	1	0.33	0.563	ns
Residual	554,050.90	6949	—	—	—

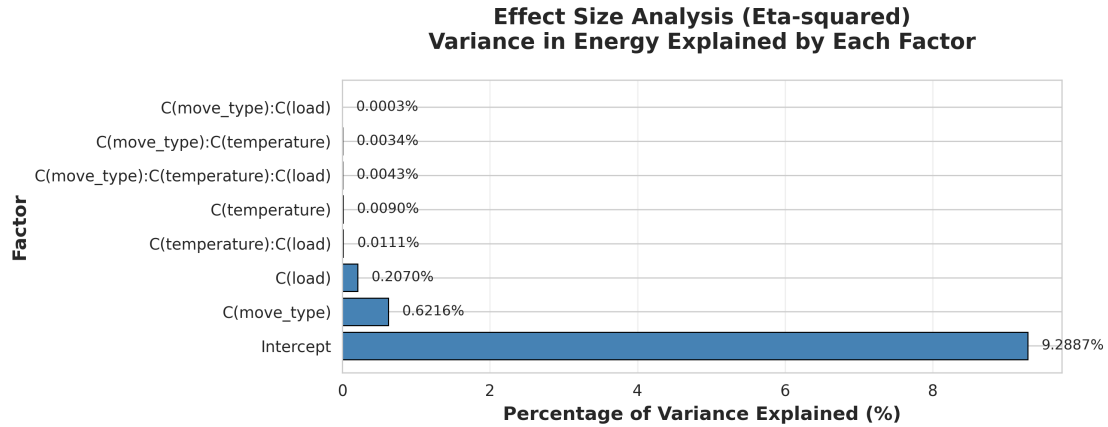


Fig. 1. Effect size analysis showing percentage of variance explained by each factor. Move Type explains 0.62% and Load explains 0.21% of the total variance in energy consumption.

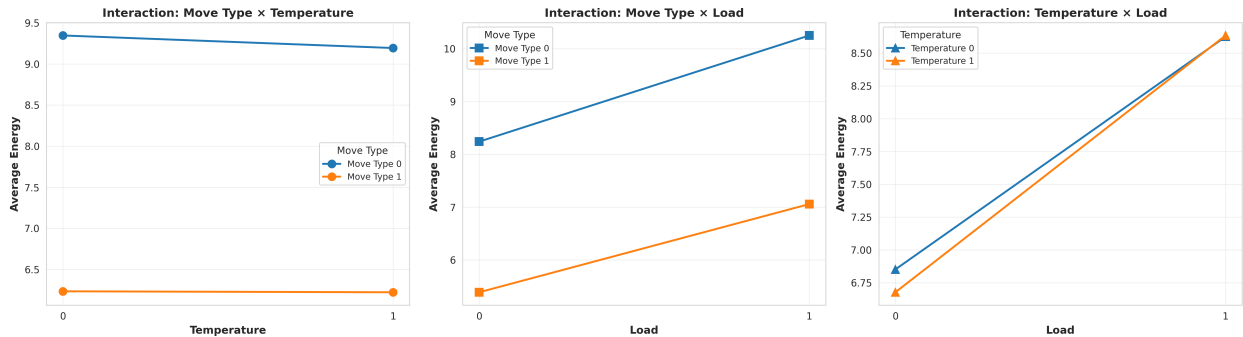


Fig. 2. Interaction plots for filtered data. (Left) Move Type × Temperature: nearly parallel lines indicate minimal interaction. (Center) Move Type × Load: both move types show similar load effects (parallel lines). (Right) Temperature × Load: parallel lines suggest additive effects with no interaction.

than Move Type = 0 ($p < 0.001$). In contrast, there is no statistically significant difference in energy consumption

between the temperature levels ($p = 0.603$). Load also has a significant effect, with Load = 0 consuming 1.87 units

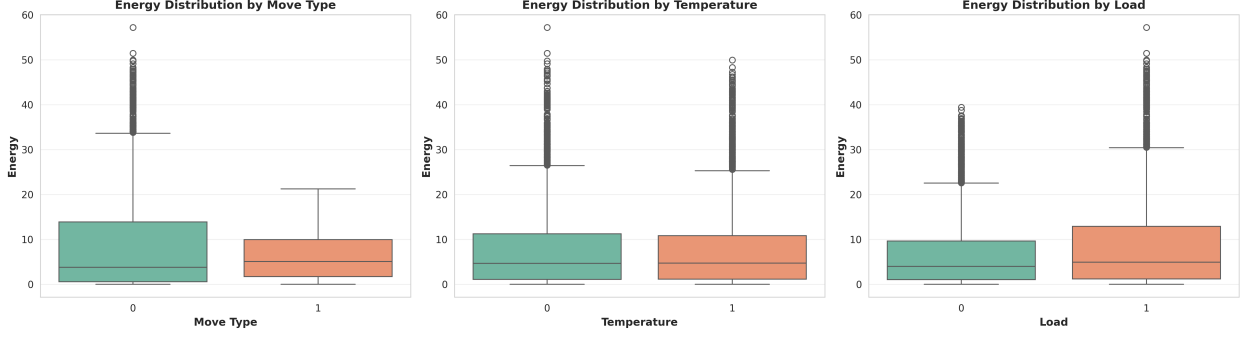


Fig. 3. Box plots showing energy distribution by factor levels in filtered data. (Left) Move Type = 0 shows higher median and mean than Move Type = 1. (Center) Temperature levels show similar distributions. (Right) Load = 1 shows higher energy than Load = 0. Note: outliers are still present but less extreme than in the full dataset.

TABLE IV
TUKEY'S HSD PAIRWISE COMPARISONS FOR MAIN EFFECTS (FILTERED DATA)

Factor	Group 1	Group 2	Mean Diff	p-adj	Lower CI	Upper CI	Reject H_0
Move Type	0	1	-3.04	<0.001	-3.47	-2.62	Yes
Temperature	0	1	-0.11	0.603	-0.54	0.31	No
Load	0	1	1.87	<0.001	1.44	2.29	Yes

less energy than Load = 1 ($p < 0.001$).

G. Random Forest Analysis

Using the aggregated mean energy for each group ($n = 8$), a Random Forest model ranked the importance of the experimental factors (Figure 4).

This confirms the ANOVA findings: Move Type is the primary high-level determinant of energy consumption, significantly outweighing Load and Temperature.

VI. CONCLUSION

Key findings are that Move Type exhibits the strongest effect on energy consumption, with Move Type = 1 consuming substantially less energy (mean = 6.2) than Move Type = 0 (mean = 9.2), corresponding to a difference of 3.04 units or a 33% reduction, and explaining 0.62% of the total variance. Load also has a statistically significant impact, with Load = 1 increasing energy consumption by 1.87 units (a 27% increase), accounting for 0.21% of the variance, and this effect remains consistent across move types, as no interaction is observed. In contrast, Temperature shows no meaningful influence on energy use, with a mean difference of only 0.11 units and a negligible contribution to explained variance. Finally, no significant interaction effects are present, as evidenced by parallel interaction plots, indicating that the effects of the factors are additive rather than multiplicative.

Based on these findings, the following energy optimization recommendations can be made:

- 1) Prioritize Move Type = 1: When possible, use Move Type = 1 to achieve 33% energy savings compared to Move Type = 0

- 2) Minimize load when feasible: Operating with Load = 0 instead of Load = 1 reduces energy consumption by 27%
- 3) Temperature is less critical: Temperature adjustments (within 0–1 range) have minimal impact on energy consumption
- 4) Combined optimization: Using Move Type = 1 with Load = 0 should yield the lowest energy consumption

A. Limitations

Despite statistical significance, the observed effects are characterized by small effect sizes, explaining less than 1% of the total variance, which indicates that other unmeasured factors play a dominant role. The generalizability of the results is limited, as the findings apply only to the binary (0/1) factor levels and cannot be extrapolated to conditions coded as -1. Furthermore, more than 99% of the variance remains unexplained, pointing to substantial residual variability that likely arises from a combination of missing explanatory variables, high measurement noise, and pronounced individual differences between observations.

This filtered analysis demonstrates that meaningful categorical effects on energy consumption exist when extreme-variance groups are excluded. In particular, Move Type exhibits a highly significant effect ($p < 0.001$), with Move Type = 1 consuming approximately 33% less energy than Move Type = 0, while Load also shows a significant influence ($p < 0.001$), with Load = 1 increasing energy consumption by roughly 27% relative to Load = 0. In contrast, Temperature does not have a statistically significant effect within the 0–1 range considered. No significant interaction effects are observed, indicating that the identified factors act independently and contribute additively to energy consumption.

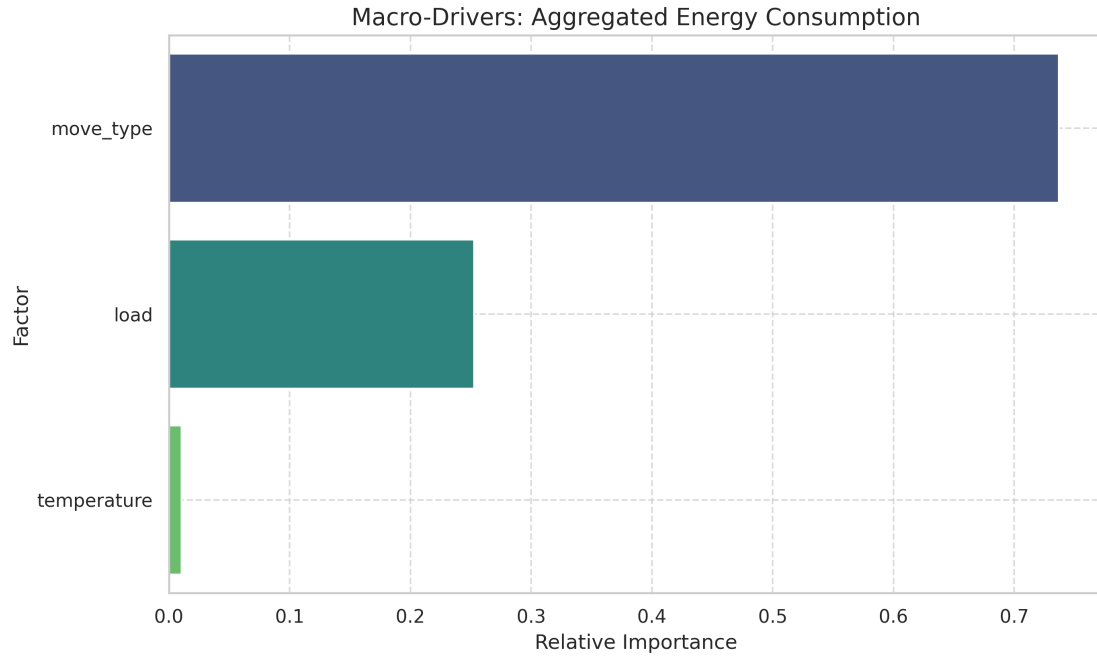


Fig. 4. Macro-level feature importance on aggregated data. Move Type is the dominant high-level factor ($\approx 74\%$), followed by Load ($\approx 25\%$). Temperature is negligible.

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